

AGRONOID: AN AGRICULTURE ROBOT

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ABSTRACT

In India, agriculture is the backbone of the economy, employing over 60 percent of the population. However, the sector faces labor shortages as interest grows in developing autonomous technologies like robots for farming. The Advance AGRONOID project seeks to address this issue through an agricultural robot designed to minimize human labor and maximize efficiency in key tasks like sowing, drilling, and soil analysis. By leveraging automation, this robot can conduct essential agricultural functions accurately and rapidly while protecting soil fertility and worker safety from hazardous conditions. The overarching goal is an autonomous, multifunction farm robot controlled through the Internet of Things to sustainably strengthen Indian agriculture.

KEYWORDS: Robot, Wireless Technology, Smartphone, Seed mapping.

1. Introduction

In our country, agriculture plays a pivotal role in bolstering the Indian economy, with a

significant portion of the rural population reliant on the agricultural sector for sustenance. The primary objective at hand is twofold: to enhance crop yield and address the persistent challenge of labour scarcity. While prior proposals have introduced various types of agricultural robots,

this latest iteration incorporates sophisticated features. The necessity for automation in agriculture primarily arises from two factors: the increasing demand for agricultural produce due to population growth and the prevailing labour shortages within the sector.

Utilizing advanced methodologies, this innovative system is capable of executing fundamental agricultural tasks while also integrating advanced functionalities such as seed sowing, drilling, watering, and analyzing soil, temperature, and humidity. The operational sequence of this robotic platform commences with soil drilling, advances to seed placement, and culminates with watering application and soil analysis. The underlying technology of the Advanced AGRONOID project leverages essential components such as DC motors, relays, Arduino nano microcontroller, and a Bluetooth module for centralized control. The communication interface for this Advanced AGRONOID is facilitated through Bluetooth connectivity.

For soil cultivation, the robot is equipped with spiked wheels affixed to its anterior end, while a seed container with a dispensing mechanism facilitates precise seed placement. Lastly, a watering unit located at the rear of the robot, controlled via relay, facilitates watering application. The overarching objective of this endeavor is to conceive a robotic solution that transcends conventional applications, heralding a new era of technological integration in agriculture.

2. Literature Survey

Suraj Chavan ,Anil kumar Dongare , Pooja Arabale , Usha suryanwanshi , Sheetal Nirve, et al., Agriculture Based Robot (AGRONOID). Many countries in Asia including India are agrarian economies and most of their rural populations depend on agriculture to earn their livelihood. Aimed at increasing the productivity and reducing the labor involved, this robot is designed to

execute the basic functions required to be carried out in farms. We aim to create a multitasking agriculture robot which will focus on basic work of plantation. To sow the seeds a robotic arm will dig to a precise depth with equal distance between the seeds. At bottom of robot water pump will be placed and as per the requirement water will be sprinkled. The right amount of fertilizer will be spread over the seed [1].

Pavan T V1, Dr. R. Suresh2, Dr. K R Prakash3, Dr. C. Mallikarjuna et al., Design and Development of Agribot for Seeding. This paper is an effort towards the automation of seeding process. The developed system is capable of performing the seeding operation in the agricultural field. The major limitation that was found in the previous systems is the seeding arrangement. The seeding arrangement is not flexible and only one type of seeds can be used. This limitation is overcome in the existing system. Based on the trials done the system is capable of performing the seeding operation. Uniform depths for the seed placement and uniform distance between the two successive seeds were obtained. The developed robot is capable of performing its operation without any human effort and the human intervention in the process is significantly reduced. Developing a robot that can perform wide variety of agricultural task is a challenging process. The use of robots can be economically justified when they are capable of performing wide variety of agricultural process without human intervention. Apart from the agricultural process these types of seeding robots can be used for mass plantation, reforestation and afforestation [2].

Mr. V. Gowrishankar, Dr. K. Venkatachalam et al., IoT Based Precision Agriculture using Agribot. More than 60 percent of the population in the India, agriculture as the primary sector occupation. In recent years, due increase in labor shortage interest has grown for the development of the autonomous vehicles

like robots in the agriculture. A robot called agribot have been designed for agricultural purposes. It is designed to minimize the labor of farmers in addition to increasing the speed and accuracy of the work. It performs the elementary functions involved in farming i.e. spraying of pesticide, sowing of seeds, and so on. Spraying pesticides especially important for the workers in the area of potentially harmful for the safety and health of the workers. This is especially important for the workers in the area of potentially harmful for the safety and health of the workers. The Proposed system aims at designing multipurpose autonomous agricultural robotic vehicle which can be controlled through IoT for seeding and spraying of pesticides. These robots are used to reduce human intervention, ensuring high yield and efficient utilization of resources [3]

Dr. Premjyoti Patil, et al., Wsn Based Advanced Agricultural Vehicle Operated Using Smart Phone – Agribot, This project introduces wireless technology in the field of agriculture. It reduces manual labor and can work in any sort of climatic condition as well as can work nonstop unlike humans. The time required to carry out the five functionalities reduces considerably in comparison with carrying out the same activities manually. Agricultural robot or AGRIBOT is a robot deployed of agricultural purposes. This multipurpose system gives an advanced method to seed sowing, ploughing, watering the crops and harvesting with minimum man power and labour making it an efficient vehicle. The whole process calculation, processing, monitoring are designed with motors and interfaced with Microcontroller [4].

Ankit Singh, Abhishek Gupta, Akash Bhosale, Sumeet Poddar. et al., Agribot: An Agriculture Robot. This paper has set out a vision of how aspects of crop production could be automated one. Although existing manned operations can be efficient over large areas there is a potential for reducing

the scale of treatments with autonomous machines that may result in even higher efficiencies. The development process may be incremental but the overall concept requires a paradigm shift in the way we think about mechanization for crop production that is based more on plant needs and novel ways of meeting them rather than modifying existing techniques. Agribot is a robot designed for agricultural purposes. As one of the trends of development on automation and intelligence of agricultural machinery in the 21st century, all kinds of agricultural robots have been researched and developed to implement a number of agricultural productions in many countries. This Bot can perform basic elementary functions like picking, harvesting, weeding, pruning, planting, grafting [5].

Ms. Aditi D. Kokate , Prof. Priyanka D. Yadav et al., Multipurpose Agricultural Robot, This project is mainly based on minimizing man power as well as cost of the equipment. The robot can be with open source system instead of normal robotic car. Automation is needed such as industry, bio-medical, survey line etc. Especially in agriculture field for increasing yield of crops. Flexibility of automation system is high than traditional system. The advantage of this system reduce the labour cost, and time. In this work a robot is built and established to carry out automatic and manual seeding, Irrigation, Fertilization in an agriculture field. The functioning of the robot is performed by renewable energy like solar energy. It is expected that the robot will support the farmers in improving the efficiency of operations in their farms [6].

3. Methodology

In this AGRONOID, there are 4 wheels, and the rear wheels are driven by a 12V DC gear motor. The seeds are sowed through the drilled hole at the bottom of the container. We use a pump sprayer to spray the pesticides. Bluetooth and Android phone is

used to control the robot for moving, plowing, drilling, and Watering. This whole robot is mainly controlled and programmed through an Arduino microcontroller which may also be known as the brain of the robot.

This Agronoid robot design, development, and evaluation adhered to a rigorous and comprehensive methodology. Hardware components were meticulously selected and designed, followed by robot assembly and control system programming. Extensive testing in controlled environments and real field's rigorously evaluated functionalities, accuracy, and economic feasibility, encompassing tasks like sowing, drilling, watering, and plowing. Sensor data, performance metrics, and cost analyses were diligently collected and meticulously analyzed, driving improvements, and comprehensively documenting the project's journey. Exclamation Safety, environmental impact, and ethical considerations were addressed throughout development. This methodology ensured a thorough evaluation of the AGRONOID's potential to transform agriculture, with findings documented in detail for future reference and potential advancements.[7]

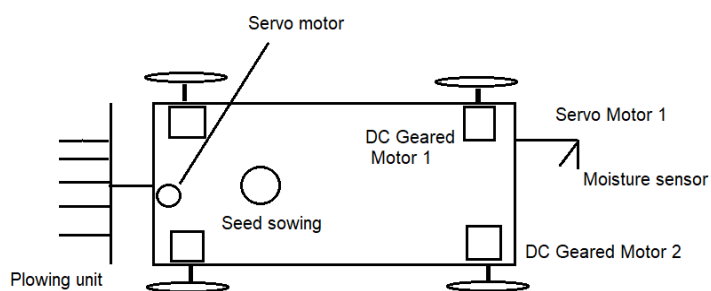


Fig 1: Structural Diagram

This multipurpose agriculture robot is used to control main three functions as we discussed above about all of them. The structural diagram of AGRONOID is given above.

4. Proposed Architecture

The deployment of the Agronoid involves

situating it on a metal sheet equipped with integrated rollers and cultivators. Positioned at the front of the sheet are harvesting features, while the cultivator end houses both a water pump for irrigation and a seed-sowing system. The movement control of the robot, encompassing forward, backward, left, and right motions, is managed by two motors. Individual motors are dedicated to operating the harvesting, seed-sowing, and watering functions. The initiation of operations is facilitated by the farmer through an Android application on their smartphone. Utilizing Bluetooth connectivity, the application serially transmits RF signals to the Agronoid's receiver Bluetooth, thereby triggering actions in accordance with the farmer's instructions. Interfacing tasks are managed by Microcontroller 8051. Powering the Agronoid are two Intel AT89C2051 microcontrollers, representing the pinnacle of Intel's 8051 family. The first microcontroller deciphers commands from the transmitter and executes the received instructions. Control over the two drive motors and all other vehicle attachments is facilitated by the LD293 motor driver IC.[8]

This is the proposed architecture which give some overview about the Advance AGRONOID:

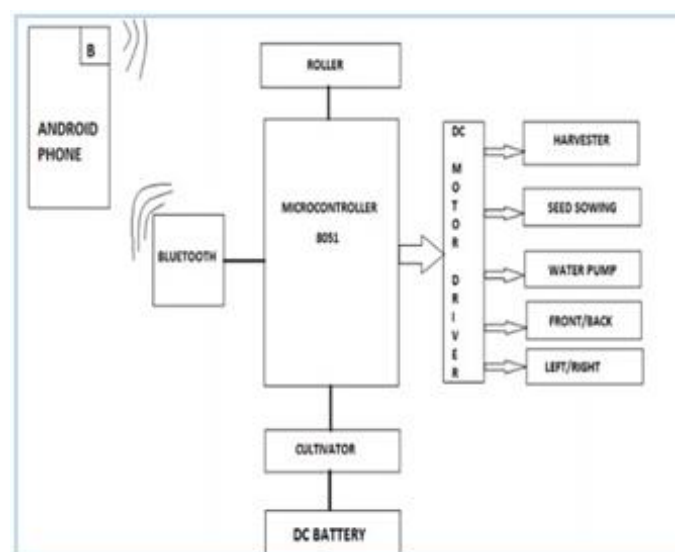


Fig 2: Architecture of AGRONOID.

DHT11 is the temperature and humidity sensor that is used to know the environmental conditions related to the temperature and humidity in the environment. When the robot starts moving in the forward direction then after a few minutes it stops and starts to drill the ground with the help of a drilling mechanism. When it is done drilling in the ground the solenoid valve starts to drop seeds in the soil.

5. Block and Circuit Diagram

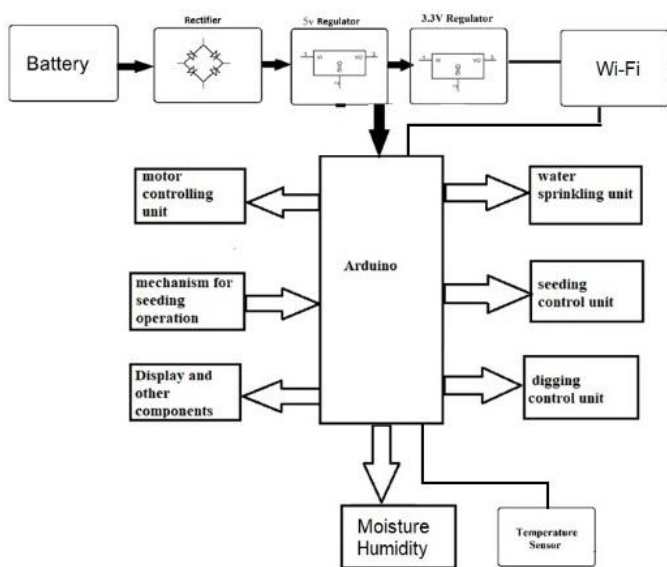


Fig 3: Block Diagram of AGRINOID.

AGRINOID is deployed on a fiber sheet developed with inbuilt roller and cultivator. Arduino will active after getting the regulated power from battery. WiFi sensor will send the signal to the Arduino for controlling the motor controlling unit, water sprinkling unit, seeding control unit, digging control unit, sensor will sense the temperature, humidity and send the signal to display unit. Seeding operation will be active after getting the signal from the control units.

Circuit diagram for Bluetooth controlled robot is shown in Fig 4 given below. A motor driver is connected to Arduino to run the robot. Motor driver's input pins 2, 7, 10 and 15 are connected to Arduino's digital pin number 12, 11, 10 and 9 respectively.

Here we have used two DC motors to driver robot in which one motor is connected at output pin of motor driver 3 and 6 and another motor is connected at 11 and 14. A 6 volt battery is also used to power the motor driver for driving motors. Bluetooth module's **rx** and **tx** pins are directly connected at **tx** and **rx** of Arduino and VCC, ground pin of Bluetooth module is connected at +5 volt and ground of Arduino. A nine volt battery is used for power the circuit at Arduino's VIN pin.[9]

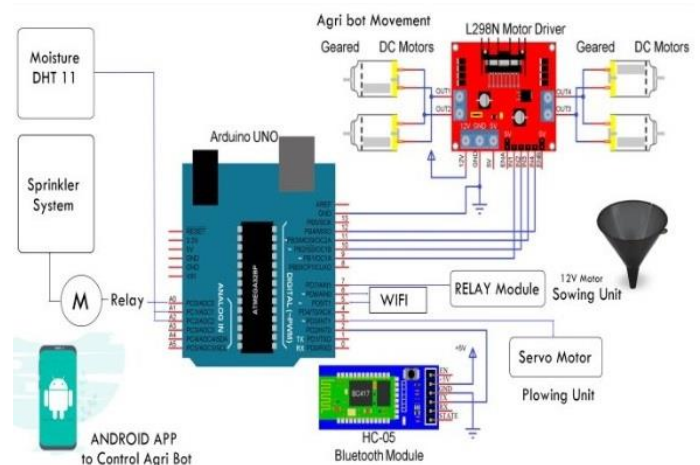


Fig 4: Circuit Diagram of AGRINOID

List of some components with specification used in this AGRINOID are as follow:

i. Arduino: The Arduino Mega 2560 is a microcontroller board based on the ATmega2560 (datasheet). It contains everything needed to support the microcontroller, simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

ii. DC Motor: DC Motor – 100RPM – 12Volts geared motors are generally a simple DC motor with a gearbox attached to it. This can be used in all-terrain robots and variety of robotic applications. 100 RPM 12V DC geared motors widely used for robotics applications. Very easy to use and available in standard size.

iii. Micro Servo Motor (SG90): It is tiny and lightweight with high output power. This servo can rotate approximately 180

degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. It comes with a 3 horns (arms) and hardware.

iv. Humidity Sensor (DHT11): A humidity sensor senses, measures and regularly reports the relative humidity in the air. It measures both moisture and air temperature. The warmer the air is, the more moisture it can hold, so relative humidity changes with fluctuations in temperature.

v. Drill Motor: An electric motor is an electrical machine converts electrical energy. The drill motor is a heavy torque motor with a drill bit connected to it.

vi. Arduino 2.2.1: The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board. Arduino IDE is a hassle-free, simple, and straightforward programming environment. With a community-driven system and simple interface, the program makes it easier to code websites and applications. You don't need to have any technical skills or knowledge to use beginner-friendly software.

Bluetooth Sensor, Soil moisture sensor, wheels, seed container, pump motor, battery etc. are also used in this robot.

6. Working of AGRONOID

This robot is deployed on a fiber sheet developed with inbuilt roller and cultivator. The front end of the fiber sheet is given the harvesting feature, while both Water pump used to water the crops and seed sowing will be added at the cultivator end. We use two motors to control the forward, backward, left and right movement. One motor each is used to control harvesting, seed sowing and watering the crops. The working begins when the Farmer opens the application and can press the options provided on the display screen. The Bluetooth on the android phone will send

the RF signals serially. We have used an L293D motor driver IC which drives two motors. These two motors are vehicle driver motors and it also runs the motors for all other attachments of agriculture in the vehicle.

The entire system is powered by a 12v battery and two voltage regulators are used to regulate voltage to 5v each for working of components. The system has a WiFi module inbuilt in it which can transfer the readings taken by the sensors and the readings can be seen in a form of graphs and charts on think speak application in real-time.[10]

7. Feature of AGRONOID

i. Ploughing Function: The initial task involves ploughing the soil, which is essential to access the nutrient-rich bottom layers. This process ensures optimal conditions for seed germination and growth, facilitating the production of nutrient-dense crops vital for sustenance and benefitting all living organisms

ii. Sowing of Seeds: Seed sowing entails the precise placement of seeds into the soil. Utilizing a container filled with seeds, the dropping mechanism is regulated by a servo motor as the AGRONOID's wheels rotate. Controlled dispensing ensures that only one seed is released at specific intervals through the container's aperture, facilitating uniform seeding and efficient crop establishment.

iii. Drilling: this advance robot is equipped with a drilling mechanism that helps to drill a hole in the soil so that seed can be sown into that hole efficiently.

iv. Watering: this advanced Agronoid also encompasses a feature of watering the seeds once the drilling is done and is used to provide moisture to the soil.

v. Soil Analysis: In this model, we are using various sensors like moisture sensor, humidity sensor, and temperature sensor which help to analyse the soil condition and try to get insights into the soil production.

vi. Robot Movement through Smartphone: This advanced AGRONOID (Agriculture Robot) is controlled by using the Bluetooth device through which we can give instructions to the robot i.e. forward, stop, right, left, and backward. In this, we use an Android version smartphone.

vii. Data visualization: The Advance robot has sensors enabled in it which help us to get insights we are using think speak to get the readings of the sensors and then showing them in the form of interactive charts and visualizations.[11]

8. Implementation states



Fig 5: Model Robot with all components



Fig 6: Temperature and Humidity Value on LCD Display

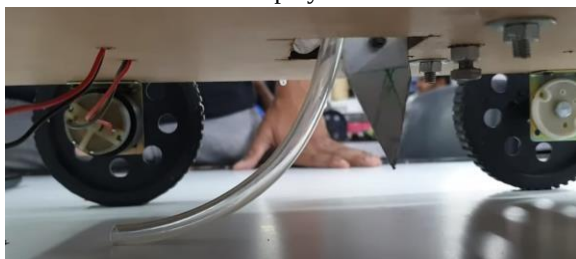


Fig 6: Drilling Unit and Watering Unit Pipe

9. Discussion

AGRONOID is driven by the goal of reducing manual labour and increasing efficiency, the robot serves as a multi-functional agricultural assistant. It boasts capabilities for autonomous seed sowing, intelligent irrigation, and smart control systems. This versatile platform offers both manual and automated operation, allowing farmers to tailor its use to their specific needs.[12]

Some key achievements come under this:

1. The robot demonstrates the ability to function independently, freeing up farmers' valuable time and resources.
2. With precise depth control, the robot ensures uniform and efficient seed distribution across the field, minimizing waste and maximizing yield potential.
3. Employing sensors, the robot gathers real-time data on temperature and humidity, enabling targeted and optimized water delivery, minimizing water usage, and promoting healthy crop growth.
4. The robot can be programmed to deliver both pesticides and water with precision, reducing manual labor and minimizing exposure to harmful chemicals for farmers.
5. By automating tasks involving hazardous chemicals and repetitive motions, this robot reduces the risk of accidents and injuries for farmers, promoting a safer work environment.

10. Conclusion

The AGRONOID, a testament to technological advancement, is poised to transform the agricultural landscape. It transcends its role as mere machinery, becoming a multi-functional partner for farmers. Its capabilities extend far beyond

line sowing, dropping seeds at precise depths, and applying pesticides with targeted precision.

This symphony of innovation delivers a resounding chorus of benefits:

- Precise sowing and targeted applications nurture healthier crops, culminating in bountiful harvests.
- By eliminating exposure to harmful chemicals, AGRONOID safeguards the health and well-being of farmers, fostering a safer work environment.
- Targeted pesticide application minimizes environmental impact, paving the way for a more sustainable future for agriculture.

AGRONOID empowers farmers to achieve greater efficiency, productivity, and sustainability, ensuring a brighter future for agriculture and the communities it sustains.

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